GeoFDI: Fault and Slip Detection as Invariance Testing, with Finite-Sample Guarantees for Legged and Wheeled-Legged Robots

Kaiwen Yang
[affiliation TBD]
[email TBD]

[advisor TBD]
[affiliation TBD]

***Abstract*—Legged robots degrade silently: an actuator loses torque, a foot slips for part of a step, an encoder drifts. Existing detectors either threshold a model residual against a distributional assumption a periodic robot violates, or learn what nominal looks like and threshold that; in both cases the false-alarm rate is empirical, and we measure it missing its target by $3\times$ to $10\times$. We reframe detection as *testing the structural invariances of the robot itself*: the mirror symmetry of the body, the space–time symmetry of the commanded gait, and the geometry of contact. Faults and slip are, by construction, breaks of these, so a group-randomisation test of the invariance hypothesis yields a false-alarm rate that is *exact at any sample size*, requires no fault data and no training, and extends to an anytime-valid sequential monitor. Because the group is explicit we can characterise what such a test can see: we prove a two-layer dichotomy separating faults that are invisible to *every* invariance test from those any consistent statistic detects, which forces a three-channel architecture and dictates when a learned nominal model must be equivariant. We further prove that a rolling contact preserves the group-affine structure of a contact-aided invariant filter, admitting wheeled-legged robots to the same machinery. Evaluation spans a mirror-exact simulator, three public datasets (Mini Cheetah, Leg-KILO Go1, Cerberus A1) and two of our own platforms. On an external fault dataset the test holds its level at 0.049 where a Mahalanobis detector fires on $52\,\%$ of healthy segments; on our own Go2, eleven sessions across three sites and two months, the recalibrated asymmetry level reproduces to within a factor 1.18; on our wheeled-legged M1 the rolling-contact filter recovers 0.99–1.00 of the driven path where the standard fixed-foot filter recovers 0.03; and as an estimator front-end the calibrated per-event gate discards $1.9\,\%$ of nominal contact measurements against $40.6\,\%$ for the standard fixed foot-speed threshold.**

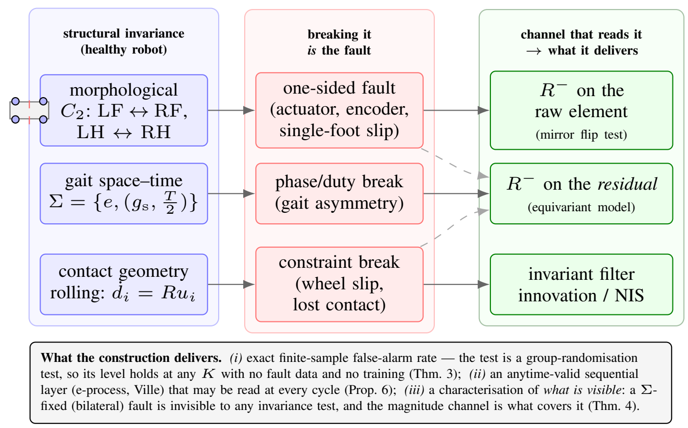

Fig. 1. **Fault and slip detection as invariance testing.** A healthy legged or wheeled-legged robot satisfies three structural invariances (left): the morphological mirror symmetry of the body, the space–time symmetry of the commanded gait, and the geometry of contact (a stance foot is fixed; a rolling wheel's contact point moves at the measured rolling rate). Every fault or slip we target *is* a break of one of them (middle), which turns detection into a test of a symmetry hypothesis rather than a comparison against a learned or thresholded nominal. Three channels read the break (right); dashed links mark the couplings that make them complementary rather than redundant.

I. Introduction

A legged robot's usefulness outside a laboratory rests on two things it cannot observe directly: whether its actuators still deliver what the controller asks, and whether its feet are where the estimator believes they are. Both degrade silently. An actuator loses torque gradually; a foot slips for a fraction of a step on a wet tile; an encoder drifts by a few milliradians. None of these announce themselves, and each corrupts the state estimate and the controller that depends on it.

The established answers fall into two families, and both leave the same hole. *Model-based* residual methods run a dynamics model — a generalised-momentum observer is the canonical choice [1], [2] — and threshold the residual, usually against a χ^2 quantile. *Data-driven* methods learn what nominal looks like, from an autoencoder or a recurrent classifier, and threshold a reconstruction error or a predicted fault parameter. The hole is the false-alarm rate. A χ^2 threshold is exact only if the residual is i.i.d. Gaussian, and a legged robot's residual is neither: it is gait-periodic, serially correlated, and non-stationary across a recording. We measure the consequence directly. On our simulation benchmark the textbook momentum-observer χ^2 detector runs at a nominal alarm rate of **0.15** against a nominal 0.05, and re-calibrating its threshold on the robot's own nominal data does not repair it — on a *held-out* stretch of the same nominal recording the rate is **0.50**, because the residual's statistics have already moved (§VI). On an external, third-party fault dataset a Mahalanobis detector fires on **52 %** of healthy segments (0.522 false alarms per healthy gap) where our test fires on **4.9 %**, i.e. at its nominal level (§VII). A detector whose false-alarm rate is unknown cannot be given authority over a controller, which is why slip handling in practice is still a hand-tuned threshold on foot speed [3].

a) Gap: Three literatures each hold one piece and none holds the combination. Morphological-symmetry work in robotics [4] uses a robot's discrete symmetry group to constrain models and policies, but treats symmetry as an architectural prior, not as a hypothesis to be tested. Fault detection and contact estimation [5], [6] build ever better nominal models and contact classifiers, but calibrate their decisions empirically. Invariance testing in statistics [7], [8] gives exactly the guarantee the robotics side lacks — a group-randomisation test whose level is exact in finite samples, and an anytime-valid sequential form — but has not been connected to a physical invariance a robot actually possesses. The missing step is to notice that a healthy legged robot *already satisfies* a group invariance precisely enough to serve as the null hypothesis, and that the faults we care about are exactly the things that break it.

b) Thesis: We propose to detect faults and slip by testing the structural invariances of the robot itself (Fig. 1). The morphological mirror symmetry of the body, the space–time symmetry of the commanded gait, and the geometry of contact are properties a healthy machine must satisfy; a one-sided actuator fault, a single-foot slip, or a broken rolling constraint are, by construction, violations of them. Framing detection this way replaces a learned or thresholded nominal with a hypothesis that needs no fault data, no training, and — in its model-free form — no dynamics model, and it inherits an exact finite-sample false-alarm rate from the group-randomisation construction.

c) Contributions:

1) **Invariance as the null, with an exact level.** We formulate the healthy-robot hypothesis H_0 as distributional invariance of a phase-registered data element under the gait symmetry group, and test it with a group-randomisation (flip) test whose false-alarm rate is *exact at any sample size*, plus an anytime-valid sequential layer built from e-processes. We also give the recalibrated null H_0' for the realistic case of a robot that is stably but measurably asymmetric, and a power lower bound with an explicit sample-size rule (§IV; Thm. 3, Prop. 5, Prop. 6).
2) **A two-layer characterisation of what is visible, and the architecture it forces.** We prove a dichotomy: a fault fixed by the symmetry group (a bilaterally symmetric degradation) is invisible to *every* invariance test, whatever its magnitude, provided the faulted loop keeps a unique symmetric attractor; anything that breaks the law is detectable by a consistent statistic. The mean-level layer gives the power of the specific paired statistic, and the two layers do not coincide — a zero-mean law change is invisible to the paired statistic and visible to an energy-distance one. This is why the system needs three channels rather than one (§IV; Thm. 4).
3) **Equivariant nominal models, and why equivariance is not optional.** When a learned dynamics model is used to form a residual, we show the residual inherits H_0 only if the model is equivariant, quantify the contamination a non-equivariant model injects, and construct an equivariant Deep Lagrangian Network by mirror weight sharing. The effect is not marginal: a plain learned model drives the residual test's false-alarm rate to 1.00 while its equivariant twin stays at 0.00 (§IV, §VI).
4) **Rolling contact preserves the invariant filter, and a five-layer evidence stack.** We prove that a rolling wheel's moving contact keeps the contact-aided invariant filter group-affine — the invariant error dynamics is unchanged, only the mean and the process noise move — which admits wheeled-legged robots to the same estimator and the same gating front-end as point-foot robots. The claims are evaluated at five levels of increasing externality, including *two* of our own hardware platforms (§VIII).

d) What the evaluation shows: On our own Unitree Go2 corpus — eleven sessions, a machine over a year in service, three sites, two dates two months apart — the naive invariance test rejects on **11/11** sessions, i.e. it sees the machine's real, stable left–right asymmetry, and the recalibrated level ν_0 reproduces across the two-month gap to within a factor **1.18**, which is smaller than the session-to-session spread within a single day. On the wheeled-legged M1 the rolling-contact filter recovers **0.99–1.00** of the driven path length where a fixed-foot filter recovers **0.03–0.04**. Used as an estimator front-end, the calibrated per-event gate discards **1.9 %** of nominal contact measurements where the standard fixed foot-speed threshold discards **40.6 %** of them on the same real outdoor data. And on the external benchmark the test holds its level at **0.049** against a nominal 0.05.

II. Related Work and Position

a) Model-based fault and collision detection: The generalised-momentum observer [1] and the family of residual generators surveyed in [2] are the reference point: they need no fault data, they isolate to a joint, and they are fast. Their decision rule is a threshold, usually a χ^2 quantile of a whitened residual, which is exact under an i.i.d. Gaussian assumption that a periodic legged system violates. We reproduce the method faithfully and measure the gap (§VI); the point of comparison in this paper is never raw sensitivity — the classical detector is often *more* sensitive — but whether the stated false-alarm rate is the delivered one.

b) Contact estimation and slip handling: Probabilistic contact estimation from foot-mounted inertial sensing [5] and learned contact classifiers [6] improve the input a state estimator receives; controllers for slippery ground [3] act on detected slip. These decide contact or slip; they do not certify how often they are wrong on healthy data. Our gate targets exactly that quantity, and we compare against the standard fixed foot-speed threshold on real outdoor data (§VIII).

c) Symmetry in robotics: Morphological symmetry groups have been used to constrain learned dynamics and control [4], and adaptation methods recover from damage without diagnosing it [9]. Both treat symmetry (or its loss) as structure to exploit; neither turns it into a hypothesis with a level.

d) Invariance testing: Group-randomisation tests give exact finite-sample validity for invariance nulls [7], and recent work provides anytime-valid e-value forms for group invariance [8], [10]. Our contribution on this side is not a new test but the identification of a physical invariance — of a robot, under its own morphology and gait — that these tests can be applied to, together with what the resulting detector can and cannot see.

Table I is the summary. Read row-wise it is not a claim that the invariance test dominates: the classical observer is faster and, at the smallest friction magnitudes, more sensitive; the GRU isolates to a joint and detects everything it was trained on. Read column-wise, the three columns that no baseline fills — a false-alarm rate that is exact rather than empirical, immunity to symmetric nuisances, and a sequential guarantee that survives continuous monitoring — are the ones a robot needs before a detector is allowed to interrupt a controller.

III. Preliminaries

a) Symmetry group and representation: Let $G = \mathbb{C}_2 = \{e, g_\mathrm{s}\}$ be the sagittal (left–right) reflection of the robot, acting on leg labels by $g_\mathrm{s} : (\mathrm{LF\,RF})(\mathrm{LH\,RH})$. Front–hind symmetry is *not* assumed: mass distribution and leg geometry differ between the pairs, so the richer $\mathbb{K}_4$ of idealised quadrupeds is unavailable. On the telemetry channel space the reflection acts by a signed permutation $\rho_\mathcal{W}(g_\mathrm{s})$: a channel maps to its mirror partner with a sign fixed by whether the quantity is polar or axial about the sagittal plane. Joint angles about the roll axis flip; pitch-axis joints do not; a foot position or velocity in the body frame is polar, so $(x, y, z) \mapsto (+x, -y, +z)$ of the partner leg; a foot force is a magnitude; the IMU accelerometer is polar and its gyroscope axial. All signs used here were verified on hardware telemetry rather than assumed from a datasheet.

b) Gait group and data element: A commanded gait adds a temporal symmetry. For a trot the relevant group is the *space–time* group $\Sigma = \{e, \sigma_*\}$ with $\sigma_* = (g_\mathrm{s}, \frac{1}{2})$: mirroring the robot and shifting by half a gait period maps the healthy motion to itself. Telemetry is phase-registered into a *data element* $Z_k \in \mathbb{R}^{d\times N}$: one gait cycle of d mirror-paired channels resampled on N phase points. For a robot without a gait phase — a wheeled-legged platform in rolling mode — the temporal factor is absent, $\Sigma = G$, and the element is a fixed-duration block. The group acts on elements by $(\rho(\sigma)Z)(\theta) = \rho_\mathcal{W}(g)\, Z(\theta + \Delta\theta)$, an orthogonal involution, so the element space splits into the isotypic components

$$\Pi^\pm = \tfrac{1}{2}(I \pm \rho), \qquad \mathcal{W} = \mathcal{W}^+ \oplus \mathcal{W}^-, \tag{1}$$

the mirror-*symmetric* and mirror-*antisymmetric* parts. $\mathcal{W}^-$ is what an invariance test can see; $\mathcal{W}^+$ is what it is structurally blind to.

c) The null hypotheses:

Definition 1 (H_0, healthy invariance)**.** $H_0 :\ \rho(\sigma)_\# P = P$ for every $\sigma \in \Sigma$, i.e. the law of the data element is invariant under the gait symmetry.

H_0 is what an ideal machine satisfies. A real machine, in service for a year and carrying a payload, is stably but measurably asymmetric, and H_0 is then false for a healthy robot — as we confirm on every one of our eleven Go2 sessions. The deployable null is therefore the recalibrated one:

Definition 2 (H_0', unchanged asymmetry)**.** Let $\nu(P) \geq 0$ be an asymmetry functional (the energy distance between $\{Z_k\}$ and $\{\rho Z_k\}$ on standardised channels), which vanishes iff the law is mirror-invariant. $H_0' :\ \nu(P_\mathrm{mon}) = \nu(P_\mathrm{cal})$ — the asymmetry *level* of the monitored window equals the level calibrated on nominal data.

H_0 is a special case ($\nu_0 = 0$). The standing assumptions behind both — equivariant dynamics, noise and command; a unique symmetric attractor; and exchangeability of elements — are stated with their ε-approximate variants and their audits in Appendix A.

IV. Theory

This section states the results and the intuition; every proof is in Appendix A, which also carries the map from the numbering here to the numbering of the technical notes.

A. An exact test, at any sample size

Theorem 3 (Exactness of the flip test)**.** *Let $Z_1, \dots, Z_K$ be data elements that are exchangeable under H_0, and let T be any statistic. Draw $M - 1$ i.i.d. uniform group elements $\sigma_m \in \Sigma$ (equivalently, i.i.d. signs on the antisymmetric component) and let $p = \frac{1}{M}\big(1 + \#\{m : T(\rho(\sigma_m)Z) \geq T(Z)\}\big)$. Then under H_0, $\mathbb{P}(p \leq \alpha) \leq \alpha$ for every K and every M, with equality when αM is an integer.*

The mechanism is that under H_0 the observed configuration and its group images are equally likely, so the rank of the observed statistic among its own orbit is uniform. Nothing is estimated, nothing is fitted, and no asymptotics are invoked: the level holds at $K = 10$ as it does at $K = 1000$. The price is that the test only sees the group — which is exactly what the next theorem characterises.

B. What is visible: a two-layer characterisation

Theorem 4 (Two-layer detectability)**.** (Layer I, law level.) *If the fault pattern is fixed by Σ (a bilaterally symmetric degradation)* and *the faulted closed loop retains a unique symmetric attractor, then $\rho_\# P_\varphi = P_\varphi$* and every *level-$\alpha$ invariance test has power $\leq \alpha$, whatever the magnitude. Conversely, if $\rho_\# P_\varphi \neq P_\varphi$, a statistic consistent for the two-sample problem between $\{Z_k\}$ and $\{\rho Z_k\}$ — the energy-distance class — has power $\to 1$.* (Layer II, mean level.) *For*

TABLE I

WHAT EACH DETECTOR REQUIRES AND WHAT IT DELIVERS. EVERY CELL IS MEASURED IN THIS PAPER; THE EXPERIMENT THAT PRODUCED IT IS NAMED. "FAR HELD" MEANS THE MEASURED NOMINAL ALARM RATE MATCHES THE STATED LEVEL.

detector	fault data	training	dynamics model	finite-sample FAR held	symmetric-nuisance immune	sequential guarantee
GeoFDI R^- (raw)	no	no	no	**yes** (0.00, e21; 0.049 ext., e03)	**yes** (0.00, e07/e21)	**yes** (Prop. 6)
GeoFDI R^- (residual)	no	no	yes	**yes** (0.00, M0.3)	**yes** (inherits H_0)	**yes**
momentum observer + χ^2	no	no	yes	**no** (0.15 fixed / 0.50 recal., e21)	no (0.90–0.95, e21)	no
Mahalanobis	no	nominal only	no	**no** (0.52 per healthy gap, e03)	no (0.50, e07)	no
autoencoder	no	nominal only	no	not stated	no (0.50, e07)	no
GRU classifier/regressor	**yes**	yes	no	empirical only (0.00–0.012, e03)	no (0.65, e07)	no

the paired-energy statistic the noncentrality after K elements is $K\|\Pi^-\mu(\varphi)\|^2_\Sigma$, and the test is consistent iff $\Pi^-\mu(\varphi) \neq 0$ *— a strict subset of the laws Layer I covers.*

Three consequences drive the system design. First, blindness to bilaterally symmetric faults is structural, not a weakness of a particular statistic, so a second, magnitude channel reading $\mathcal{W}^+$ is *necessary*, and it cannot inherit the same nuisance immunity. Second, the two layers differ exactly on zero-mean law changes: a one-sided variance inflation has $\Pi^-\mu = 0$ but breaks the law, so the paired statistic is blind while the energy-distance one is consistent — deployment runs both. Third, the blindness is robust in amplitude: sweeping a bilateral fault down to $30\,\%$ of nominal torque left the test at its level and the realised gait's chirality at its nominal floor, so the escape route is a genuine symmetry-breaking bifurcation of the gait, not a large enough fault.

Proposition 5 (Power and sample size)**.** *Under a Gaussian model for the differenced scores, the power of the whitened quadratic flip statistic at level α is at least $1-\exp\big(-(d+\lambda-c_\alpha)^2_+/(4(d+2\lambda))\big)$ with $\lambda = K\|\Pi^-\mu\|^2_\Sigma$, which inverts to an explicit sample-size rule $K \geq \lambda^*(d,\alpha,\beta)/\|\Pi^-\mu\|^2_\Sigma$.*

C. Reading it every cycle

Proposition 6 (Anytime-valid sequential layer)**.** *Let $p_1, p_2, \ldots$ be the window p-values of Thm. 3 and $e_t = \frac{1}{2}p_t^{-1/2}$. Then $E_t = \prod_{s\leq t} e_s$ is a nonnegative supermartingale with $E_0 = 1$, and by Ville's inequality the rule $E_t \geq 1/\alpha$ has false-alarm probability at most α over an unbounded horizon.*

This is what makes the test deployable: it may be inspected after every cycle, and stopping when it looks interesting does not invalidate it.

D. Residual channels and equivariance

Running the same test on the residual of a nominal dynamics model, $r = y - \hat{f}(x,u)$, both sharpens it and endangers it.

Proposition 7 (Residual inheritance and its contamination)**.** *If the nominal model $\hat{f}$ is Σ-equivariant, the residual element inherits H_0 exactly, and the flip test on it retains its level while gaining power against faults that act through the dynamics. If $\hat{f}$ is not equivariant, its equivariance defect enters the residual as a fixed asymmetric offset and inflates the level.*

The effect is large enough to matter in practice, not merely in principle: on the welded-leg world a plain learned model drives the residual test's size to 1.00 while its equivariant twin sits at 0.00 (§VI). Mirror weight sharing is what makes a Deep Lagrangian Network equivariant by construction, and the equivariance defect is then exactly zero rather than small.

E. Rolling contact keeps the invariant filter

Proposition 8 (Rolling contact is group-affine)**.** *For a wheel in contact with rolling constraint $\dot{d}_i = R\,u_i$, u_i a measured input, the contact-aided invariant filter's propagation remains group-affine, and the right-invariant error dynamics of the contact block is* unchanged*: $A[d_i, R] = 0$, exactly as for a world-fixed foot. Rolling alters only the mean propagation $d_i \mapsto d_i + Ru_i\Delta t$ and the contact process noise, which becomes anisotropic in the wheel frame.*

The plain-EKF Jacobian for a moving world point is $-R(u_i)_\times$, and carrying it into an invariant filter is a natural and wrong step that makes the error dynamics state-dependent. The correct statement admits wheeled-legged robots to the same estimator, and hence the same gating front-end, as point-foot robots. The signature of a fault in such a filter is the observable projection of its direction transported by the adjoint, which is why a bilaterally symmetric slip is invisible to the invariance channel but visible to the estimator channel — the property that turns the pair into a slip *classifier* (§V).

V. METHOD

A. Three channels

The detector is three readings of the same telemetry, forced by Thm. 4 rather than assembled by taste. R^- tests invariance on the raw element and sees exactly the antisymmetric component $\mathcal{W}^-$: one-sided faults and one-sided slip, with an exact level and no model. R^+ reads the symmetric component $\mathcal{W}^+$ of a model residual and covers what R^- is structurally blind to — bilaterally symmetric degradation — at the cost of needing a dynamics model and of inheriting no nuisance immunity. The estimator channel reads the innovation of the contact-aided invariant filter, which responds to constraint violations (slip, lost contact) whether or not they are symmetric. Running R^- on the *residual* rather than the raw element is a fourth combination and is what makes friction-type faults visible to the invariance test at all (§VI).

Algorithm 1 GeoFDI tester (one session)

1: **segment**: cut telemetry into steady, straight, single-condition runs
2: **register**: estimate gait phase; resample each cycle onto N points $\to$ elements Z_k (rolling mode: fixed-duration blocks)
3: **calibrate**: on the first K_{cal} elements of *this run*, estimate the asymmetry level ν_0 and the channel scales
4: **test**: per window of w elements, flip test (Thm. 3) with statistics {paired-energy, energy-distance} $\to p_t$
5: **accumulate**: $e_t = \frac{1}{2} p_t^{-1/2}$, $E_t = \prod_{s \le t} e_s$; alarm if $E_t \ge 1/\alpha$ (Prop. 6)
6: **recalibrate**: on hardware, replace H_0 by H_0' — test $\nu_{\mathrm{mon}} = \nu_{\mathrm{cal}}$
7: **isolate**: rank mirror *pairs* by Π^- energy; obtain the *side* from the signed mean
8: **gate** (optional): per-stance-event e-values $\to$ per-leg weights $\pi_i \to$ invariant-filter update

TABLE II
THE NINE CONDITIONS ALGORITHM 1 DEPENDS ON, AND WHERE EACH IS ESTABLISHED.

#	condition	established by
C1	the plant, command and noise are Σ-equivariant	assumptions A1–A4, audited
C2	the healthy loop has a *unique* symmetric attractor	A5; falsified by a chiral gait at low gain
C3	elements are exchangeable across the index	Assumption E; block length $\ge 2\times$ the nuisance correlation time
C4	the phase registration is half-period equivariant	$\pm 10\,\%$ clock error costs power, not level
C5	the segment is a single homogeneous condition	per-run calibration (§V-C)
C6	on hardware, use H_0': the level ν_0, not $\nu = 0$	Def. 2; 11/11 sessions reject H_0
C7	a residual channel needs an *equivariant* model	Prop. 7; plain model $\to$ size 1.00
C8	bilaterally symmetric faults need the R^+ channel	Thm. 4 (Layer I)
C9	**side attribution needs a *signed* statistic**	energy statistics rank mirror *pairs* only

B. The tester

Algorithm 1 is the whole procedure. Its correctness rests on nine conditions, listed beside it; each is either a theorem hypothesis or an empirical calibration, and each is tagged with where it is established.

C9 deserves emphasis because it is easy to get wrong and we did get it wrong once. For a mirror pair (c, c') the antisymmetric component satisfies $(\Pi^- Z)[c'] = -s\,(\Pi^- Z)[c]$, so the two partners carry *identical* Π^- energy: no energy-type statistic can say which side is at fault, only which pair. Standardising each channel by its own scale before summing breaks that identity numerically and manufactures a spurious side ranking. The side is recoverable only from the *sign* of the mean difference, which is what Algorithm 1 line 7 uses.

C. Calibrate per traverse, not per session

The single most consequential deployment rule we found is not in the theory but in the hardware data. On both of our platforms the recalibrated null H_0' holds *within* a continuous traverse and fails *across* traverses: on the Go2 corpus the sequential monitor alarms on 8 of 11 sessions when the element pools several straight runs, but on only 2 of 11 when it is built from a single run, and a shape classifier labels 8 of 11 ν-trajectories as boundary-jumps and none as drift. The same signature appears on the wheeled M1. The mechanism is that the healthy asymmetry level depends on heading, ground and payload placement, so concatenating traverses violates the premise that calibration and monitoring windows come from one regime. The rule is therefore: *calibrate ν_0 per homogeneous segment*. A segment is not automatically homogeneous merely because it is one continuous run — §VIII reports a session where one spatial half is internally clean and the other is not.

D. Equivariant nominal model and π_i gating

The learned nominal model is a Deep Lagrangian Network made equivariant by mirror weight sharing: one template per leg pair, applied to the mirrored coordinates of the partner leg, so the equivariance defect is exactly zero rather than small. For the estimation front-end, each stance event (a touch-down and the contact interval it opens) is scored by the pre-update per-foot innovation, converted to a conformal p-value against a library of nominal events, and turned into an e-value; the per-leg e-values become weights π_i that either drop (hard gate, $e \ge 1/\alpha$) or down-weight (soft gate, covariance scaled by $1/w_i$) that foot's measurement. Because the gate is calibrated rather than thresholded, its per-event false-alarm rate is $\approx \alpha$ by construction — which is the property the standard fixed foot-speed threshold lacks (§VIII).

E. Implementation and compute

The detector is deliberately cheap: the flip test is a few hundred signed sums over a 34×64 element. Timing every stage on recorded sessions, on a single pinned core with BLAS restricted to one thread, gives 3.4 ms of detector work per gait cycle on the Go2 against a 444 ms cycle period (a factor 131), and 2.3 ms per one-second block on the wheeled M1 (a factor 444). The cost is dominated by the gait-phase estimate (1.47 ms) and the H_0' permutation test (1.75 ms); the flip test itself costs 0.06 ms at $M = 512$. The contact-aided invariant filter, propagate plus update, runs at 0.163 ms per telemetry sample, i.e. a sustainable 6.1 kHz against a 250 Hz telemetry rate. *These are laptop-class numbers* (`Intel Core Ultra 9 275HX`) and are reported as a proxy for onboard compute; the same script and CSV schema are set up to be re-run on an onboard-class machine (`docs/protocol/bench_nuc.md`).

VI. Simulation Study

The simulation layer isolates the statistical claims from modelling error: the world is mirror-exact by construction, so a level that fails here is the test's fault and not the robot's. All runs use a MuJoCo Unitree Go2 with a symmetrised model, $\alpha = 0.05$, and deterministic seeds.

a) The level holds, and what breaks it: Under H_0 the flip test sits inside its nominal band across process-noise levels, and the three standard robustness probes behave as follows. A systematic gait-phase clock error of ± 2, ± 5, $\pm 10\,\%$ leaves the size at 0.000–0.050, i.e. inside the band at every point: a misregistered phase *decorrelates* the mirror pairing and makes the test conservative, so it costs power, not level. The H_0' calibration set may be small: $K_{\text{cal}} \in \{60, 200, 400\}$ gives sizes 0.083/0.033/0.000, all in band, so 60 nominal cycles already suffice for validity. The one probe that does break the level is the block length: under an autocorrelated but mirror-symmetric-in-law nuisance with correlation time τ, flip blocks of 0.5τ, 1τ, 2τ give sizes 0.350, 0.217, 0.100. *The block must be at least twice the nuisance correlation time*; this is the single number a practitioner must get right, and it is condition C3 of Table II.

b) Isotypic prediction: The two-layer theorem is directly visible. A single-leg fault splits its footprint exactly half-and-half between $\mathcal{W}^+$ and $\mathcal{W}^-$; a bilaterally symmetric fault of equal magnitude puts all of it in $\mathcal{W}^+$ and R^- alarms at 0.00 while the magnitude channel alarms at 1.00; an unequal bilateral pair lands in between. Sweeping the symmetric fault down to $30\,\%$ of nominal torque leaves R^- at its level throughout, with the realised gait's chirality index pinned at its nominal floor — the blindness is robust in amplitude, and its escape route is a bifurcation of the gait, not a bigger fault. The statistic split predicted by Layer I versus Layer II is also confirmed: against a zero-mean variance change the paired statistic stays pinned at α while the energy-distance statistic rises to 0.88.

c) Equivariance is not optional: On the welded-leg world, the flip test on a residual formed with an *equivariant* learned model has size 0.00 and with the analytic model 0.02, both inside the band; formed with a plain (non-equivariant) learned model of the same architecture and training budget, the size is 1.00. The recalibrated H_0' construction restores it to 0.04, while naively subtracting the calibration mean does not (it stays at 1.00) — centring is not a substitute for the differenced null.

d) Against the classical baseline, on identical rollouts: Table I summarises; the numbers are these. The momentum-observer χ^2 detector attains mean detection 0.77 at its textbook threshold and 0.96 with a horizon-recalibrated one, against 0.71 for the raw-element R^-, and it is faster (median delay 4.6–13.5 cycles versus 25). But its measured nominal alarm rate is 0.15 and 0.50 respectively, against 0.00 for R^-, so those detection numbers are not at equal false-alarm rate. Under a symmetric torque/friction drift — a nuisance, not a fault — the classical detector alarms on 0.90–0.95 of runs and the autoencoder and Mahalanobis detectors on 0.50, while R^- stays at 0.00.

e) Friction, and which channel sees it: The one fault family where the raw-element R^- is genuinely blind is friction. Moving the same test onto the analytic residual closes it: at `friction_scale` $\times 1.5$ the raw element detects 0.00 and the residual element 1.00, both at a measured nominal alarm rate of 0.00, where the classical detector reaches 0.85 only at a false-alarm rate of 0.50. Below that, at $\times 1.2$, no detector catches the fault while holding its stated level. This is the empirical content of Prop. 7 and the reason the architecture carries a residual channel rather than only a raw one.

f) Slip as a classification problem: Finally, the two-channel structure separates slip *regimes* rather than merely detecting slip. With a one-sided low-friction patch, R^- reaches power 1.00 while the estimator's innovation statistic stays flat; with uniformly low friction under all four feet, R^- stays at α — as Thm. 4 requires — and the estimator's NIS ratio rises to 1.42. Reading the pair therefore tells unilateral from bilateral slip, which a single channel cannot do.

VII. External Benchmarks

Three third-party datasets test the claims on data we did not generate, on three robots we do not own.

a) A fault dataset with labels: On the Liu et al. A1 actuator-fault dataset, the half-cycle R^- test with its sequential layer detects 0.73/0.84/0.75/0.68 of single, same-side, mirrored and diagonal double faults respectively, at **0.049 false alarms per healthy gap** against a nominal 0.05 — i.e. the exactness claim survives contact with foreign data. The comparison detectors on the same episodes are informative in both directions: a GRU trained on this dataset's own faults detects essentially everything (≈ 1.00) and isolates to a joint, which a test needing no fault data cannot match; a Mahalanobis detector matches its detection but fires on **0.522** of healthy gaps, a $10\times$ higher false-alarm burden than ours at comparable sensitivity. The pre-registered blindness check — the mirrored double fault, which theory says must be invisible — is *inconclusive* here: the straight-segment subset contains only $n = 8$ such episodes and returns 0.50, too few to separate blindness from low power. We report it as a gap rather than as support.

b) Cross-terrain breadth: The MIT Mini Cheetah contact dataset gives eight terrains at 1 kHz with labelled contacts. The detection channel fires on **all eight** (naive H_0, $p \approx 0.002$ everywhere): the machine's real mirror asymmetry is visible regardless of ground. The recalibrated H_0' layer, however, is *also* elevated on this corpus, and we could not rescue it: replacing cycle registration by rolling-style time blocks with a phase-free element reduced the sequential alarms from 6/8 to 2/8, but that reduction is confounded by the block construction leaving only 1–4 monitoring windows instead of 7–20, and the window-reject *rate* barely moves ($0.397 \to 0.375$). We therefore record an honest scope limit: on a 0.25 s flying trot with a flight phase the healthy asymmetry level is not

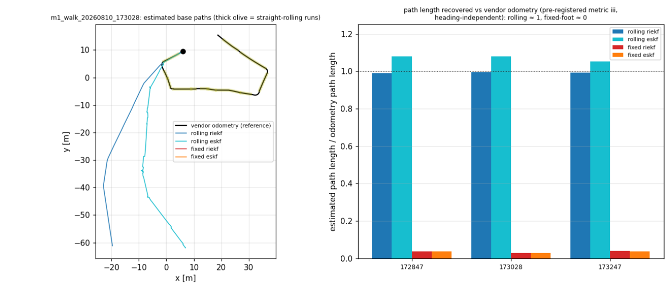

Fig. 2. Wheeled-legged M1, real rolling traverses. Left: estimated base paths against the vendor odometry (thick olive = the straight-rolling runs used for testing). Right: the heading-independent metric — fraction of the driven path length recovered. The rolling-contact model recovers essentially all of it; the fixed-foot model recovers almost none, because it treats a continuously rolling robot as near-stationary.

stationary at the resolution these segments allow, and H_0' needs a slower, well-registered gait.

c) Two more platforms: On the Leg-KILO Go1 sequences the picture is the clean one: naive H_0 rejects on every sequence, and H_0' is *in band* on 4 of 5 (the fifth is a 21-cycle fragment). On the Cerberus Street A1 sequence naive H_0 rejects and H_0' is partly elevated (0.29) over a 260 m outdoor walk. Together with the Mini Cheetah result this bounds the H_0' layer's applicability empirically: it holds on walking trots and rolling gaits, and not on aggressive ones.

VIII. Hardware Evaluation

Two of our own robots, recorded independently of this work, provide the evidence that matters most: a wheeled-legged M1 and a Unitree Go2 that has been in service for over a year.

A. Wheeled-legged M1: rolling contact

Four sessions were recorded on the M1 (16 joints, wheel-footed), of which three are rolling traverses with a vendor odometry reference. Two results follow.

First, the rolling-contact filter of Prop. 8 is not a refinement but the difference between working and not working. Measured against the odometry, the rolling filter recovers **0.99–1.00** of the driven path length on the three sessions, and a fixed-foot contact-aided filter — the standard construction — recovers **0.03–0.04** (Fig. 2); per straight run the travelled-distance error is 0.6–1.4 % against 98 %. The classical-error-coordinate version of the same rolling model performs equivalently, confirming that the gain comes from the contact model rather than from the error parametrisation.

Second, the invariance layer behaves exactly as H_0' predicts for a real machine: the naive test rejects on all three rolling sessions ($p \approx 0.002$), i.e. the robot has a real and stable asymmetry, while the sequential H_0' monitor stays silent within each session. One session is the exception that produced the per-traverse rule of §V-C, and is discussed with the Go2 case below.

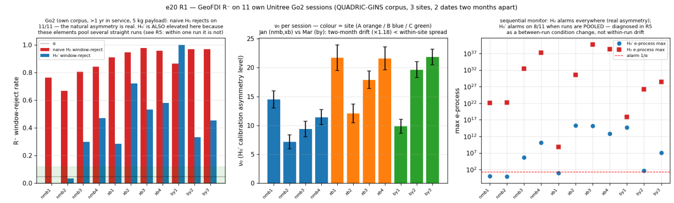

Fig. 3. Own Go2 corpus, eleven sessions. Left: the naive invariance test rejects everywhere — the natural asymmetry of a machine over a year in service. Middle: the recalibrated level ν_0 per session, coloured by site; January (`nmb`, `xb`) versus March (`by`) is the two-month reproducibility check. Right: sequential monitors.

B. Unitree Go2: eleven sessions, three sites, two months

The Go2 corpus is eleven sessions recorded on two dates two months apart, at three sites (an indoor–outdoor tile transition, a GNSS-restricted smooth area, and rough ground; site labels confirmed by the operator), at ≈ 1 m/s under a constant ≈ 5 kg payload, with a Fixposition Vision-RTK2 reference. The recorded interface is the vendor's high-level API, which carries *no* joint stream; the mirror element is therefore built from the robot's own forward kinematics — body-frame foot positions and velocities — plus foot forces and the IMU, 34 channels of which 28 are leg-resolved.

a) The asymmetry is real, large, and reproducible: The naive test rejects on **11/11** sessions (Fig. 3), with per-window rejection 0.67–0.98: a machine of this age carrying a payload is unambiguously not mirror-symmetric, which is precisely why H_0' rather than H_0 is the deployable null. The recalibrated level ν_0 ranges over 7.2–21.9, and **across the two-month gap its mean changes by a factor** 1.18**, smaller than the session-to-session spread within a single site and day** (standard deviations 3.1–6.4). The asymmetry level of this robot is thus a reproducible property. An H_0' *calibration*, however, does not transfer: applying one session's calibration to another gives window-rejection 0.32–0.49, far above α, and same-site pairs (0.33) are barely better than different-site pairs (0.37).

b) Where the asymmetry lives, and what does not *cause it:* Because R^- ranks mirror pairs and the side comes from the signed mean (condition C9), the localisation is reported both ways. The hind pair carries the larger antisymmetric energy share in 6 of 11 sessions, and the signed statistic is remarkably stable: **in 11/11 sessions the left-hind minus right-hind foot force is negative and the left-front minus right-front is positive** — one consistent diagonal loading pattern across three sites and two months. A foot-mounted inertial board on the left-hind leg was the obvious candidate cause; the March sessions carry *no such board* and show the same or a larger asymmetry, so the board is ruled out. The remaining candidates are payload placement and accumulated wear, which a payload-swap session would separate.

c) Residual anomalies, and the limit of the per-traverse rule: The per-traverse calibration rule of §V-C came from this corpus: pooling straight runs alarms on 8/11 sessions and single runs on 2/11, with 8/11 ν-trajectories classified as boundary jumps and none as drift. We then split the two sessions that alarmed *inside* a single traverse. Neither pre-registered outcome occurred. In both, one half is clean and the other is not: for one session the spatial half beyond the track midpoint has window-rejection 0.00 with a sequential statistic of 0.54 (nowhere near the $1/\alpha = 20$ alarm line) while the other half alarms; for the second session the first temporal half is in band (0.07) and the second alarms. The inhomogeneity is therefore real and *localised* rather than an artefact of joining runs. For the first session the symmetric readouts across the split are identical (foot-force level differs by 1 %, high-frequency IMU texture by 3 %), so the split does *not* correspond to the site's tile boundary and the surface-switch explanation is unsupported; for the second, foot-force level moves by 17 % and the heavier half is the one that alarms. These two segments are carried as *residual anomalies*: reproducible rejections on a healthy machine whose cause the recorded channels do not identify.

d) The gate on real outdoor data: The high-level API publishes body-frame foot positions, which is exactly what a contact-aided invariant filter consumes, so the estimator experiment ran despite the absent joint stream. Trajectory error against the quality-gated RTK reference is $\approx 6.6\,\%$ of path length and is *unchanged* by gating — these traverses contain no slip severe enough to repair, and we report that null. What does differ is the false-alarm burden: **the standard fixed 0.4 m/s foot-speed threshold discards 40.6 % of nominal contact measurements, the calibrated per-event gate** 1.9 %, on the same data. This is the real-robot form of the guarantee, and it is the one property that survives the absence of a severe slip event.

Injection experiments not yet run (the robot day). The evaluation above is entirely *observational*: no fault was injected on either of our platforms. The following are pre-specified and will fill the correspondingly named rows; each is listed so that a reader can see exactly what is missing.

- (P1) **Unilateral low-friction patch** (PTFE/acrylic under one track) — fills the one-sided slip row of the slip-classifier table with hardware, against the simulated R^- power 1.00.
- (P2) **Bilateral low-friction patch** (full-width) — the blindness prediction on hardware: R^- at α, estimator NIS elevated.
- (P3) **Unilateral ballast** — a controlled one-sided mass asymmetry of known magnitude, to calibrate the detectable-magnitude axis against the simulated minimum-detectable table.
- (P4) **Payload diagonal rotation** — rotating the 5 kg payload between diagonal placements, to separate payload offset from wear as the cause of the stable LH−RH / LF−RF pattern reported above.
- (P5) **Continuous straight runs $\geq$ 60 s (M1)** — the current runs are 12–14 s, too short for a within-run H_0' control; this is what would settle the residual-anomaly question on that platform.
- (P6) **`LowState` recording (Go2)** — the joint/torque stream the high-level API cannot provide; unlocks the residual (R^+) channel and the friction-detectable row on this robot.
- (P7) **Joint-level command access (M1)** — required for controlled actuator-fault injection and for the commanded-torque residual.

IX. Limitations

We list what the construction cannot do, in the order a sceptical reader will ask.

1) **The invariance channel sees only the antisymmetric component.** This is structural (Thm. 4), not a property of our statistic: a bilaterally symmetric degradation is invisible to *any* invariance test at *any* magnitude while the symmetric attractor persists. The magnitude channel covers it, and that channel inherits none of the nuisance immunity — a symmetric drift inflates it (measured: 0.50–0.95 alarm on a nuisance where R^- is at 0.00). The guarantee is therefore asymmetric across the two channels by construction.
2) **The mirrored double fault remains empirically open on external data.** Our own simulations confirm the blindness, but the one third-party dataset containing mirrored double faults offers $n = 8$ straight-segment episodes, which is too few to distinguish blindness from low power. The clean external demonstration is still owed.
3) H_0' **fails on aggressive gaits.** On a 0.25 s flying trot the recalibrated level is not stationary at the resolution the data allow, and replacing phase registration by time blocking does not rescue it (the apparent improvement is confounded by window count). The layer is validated on walking trots and rolling gaits only.
4) **An H_0' calibration does not transfer across traverses.** The *level* ν_0 reproduces over two months, but a calibration taken from one run applied to another gives window-rejection 0.32–0.49. Deployment must calibrate per homogeneous segment, and we show a session where one continuous run is itself not homogeneous.
5) **Site-A's surface switch is not resolved.** We could not certify from the recordings which half of a session was indoors, and the split we could define does not move the symmetric readouts. The surface-switch interpretation is an interpretation, not a measurement.
6) **The model-free variant cannot see friction.** Friction faults require the residual channel and hence a dynamics model; the raw-element test detects none of them at $\times 1.5$ while holding its level. Where no model is available, that fault family is out of scope.
7) **No fault was injected on our own robots.** All hardware evidence is observational (§VIII); the injection experiments are enumerated in the placeholder box and are the paper's largest open item.

8) **The onboard numbers are a proxy.** Timing is measured on a laptop-class CPU pinned to one core, not on the robot's own computer; §V reports it as such.

X. Conclusion

We proposed treating fault and slip detection as a test of the invariances a healthy robot must satisfy — the mirror symmetry of its body, the space–time symmetry of its gait, and the geometry of its contacts. The reframing buys a false-alarm rate that is exact in finite samples with no fault data and no training, an anytime-valid sequential form, and — because the group is explicit — a precise statement of which faults are invisible and what second channel is needed to cover them. It also transfers to wheeled-legged machines: a rolling contact leaves the invariant filter's error dynamics untouched, so the same estimator and the same gate apply.

The evidence runs from a mirror-exact simulator through three public datasets to two of our own robots, and the hardware is where the reframing earns its keep: on a Go2 over a year in service, the recalibrated asymmetry level reproduces across two months to within a factor 1.18, and used as an estimator front-end the calibrated gate discards $1.9\,\%$ of nominal contact measurements where the standard threshold discards $40.6\,\%$. What remains is injection: every hardware number here is observational, and the enumerated experiments would convert the detectability statements from simulated to measured.

The longer aim is that π_i — a per-leg, per-event, calibrated trust weight — becomes the standard front-end between a legged robot's contacts and everything downstream that assumes they are trustworthy.

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Appendix A
Proofs and Numbering Map

The technical notes behind this paper are organised in four parts; the map below relates the numbering used in the body to the numbering there, so a reader can follow a proof to its full statement with its assumptions and audits.

TABLE III
Numbering map, manuscript $\rightarrow$ technical notes.

manuscript	technical notes	workstream
Def. 1 (H_0)	Part 0, Def. "h0"	—
Def. 2 (H_0')	Part 0, Def. "h0prime"	—
Thm. 3	Part 1, Thm. N1-1	N1-1
Thm. 4	Part 1, Thm. N1-2 (two-layer)	N1-2
Prop. 5	Part 1, Prop. N1-3	N1-3
Prop. 6	Part 1, Prop. N1-5	N1-5
Prop. 7	Part 2, Prop. N3-1 / N3-2	N3
Prop. 8	Part 3, rolling group-affine	N2
Table II C1–C3	Part 0, A1–A5 and Assumption E	—

A.1 Exactness (Thm. 3)

Under H_0 the joint law of $(Z_1, \dots, Z_K)$ is invariant under the group action applied elementwise, so for a random σ drawn uniformly the pair $(T(Z), T(\rho(\sigma)Z))$ is exchangeable. The rank of $T(Z)$ among $\{T(Z)\} \cup \{T(\rho(\sigma_m)Z)\}_{m<M}$ is therefore uniform on $\{1, \dots, M\}$, and the Monte-Carlo p-value $p = (1 + \#\{T(\rho(\sigma_m)Z) \geq T(Z)\})/M$ satisfies $\mathbb{P}(p \leq \alpha) = \lfloor \alpha M \rfloor / M \leq \alpha$. No distributional assumption on Z enters, which is why the level is exact at every K. □

A.2 Two-layer detectability (Thm. 4)

Layer I. A Σ-fixed fault maps the closed loop to itself, so equivariance alone gives that $\rho_{\#}\mu$ is invariant whenever μ is; uniqueness of the invariant measure for the *faulted* loop then forces $\rho_{\#}P_\varphi = P_\varphi$, and Thm. 3 makes every flip test level-α under P_φ. Conversely if $\rho_{\#}P_\varphi \neq P_\varphi$ the two laws differ, some bounded continuous h separates them, and the energy distance (equivalently a characteristic-kernel MMD) is zero iff the laws coincide and is estimated $\sqrt{K}$-consistently, so the permutation test built on it has power $\to 1$. *Layer II.* With $D_k = Z_k - \rho Z_k$ we have $D_k = 2\Pi^- \mu + (\xi_k - \rho\xi_k)$, whence $\mathbb{E}\|\bar{D}\|^2 = 4\|\Pi^-\mu\|^2 + \operatorname{tr}\operatorname{Var}(D_1)/K$ against a flip-null expectation of $\operatorname{tr}\operatorname{Var}(D_1)/K + O(\|\Pi^-\mu\|^2/K)$; consistency follows by the law of large numbers and holds iff $\Pi^-\mu \neq 0$. Since $\Pi^-\mu = 0$ is possible with $\rho_{\#}P_\varphi \neq P_\varphi$ (equal means, unequal higher moments), Layer II covers a strict subset of Layer I. □

A.3 Power bound (Prop. 5)

The whitened statistic is $\chi^2_d(\lambda)$ with $\lambda = K\|\Pi^-\mu\|^2_\Sigma$. Laurent–Massart's lower tail gives $\mathbb{P}(T \leq d + \lambda - 2\sqrt{(d+2\lambda)x}) \leq e^{-x}$; choosing x so the endpoint equals c_α yields the stated bound, and requiring $x \geq \ln(1/\beta)$ together with the companion upper tail for c_α gives the sample-size rule after solving a quadratic in $\sqrt{\lambda}$. □

A.4 Sequential validity (Prop. 6)

For a super-uniform p, $\mathbb{E}[\frac{1}{2}p^{-1/2}] \leq \frac{1}{2}\int_0^1 u^{-1/2}du = 1$, so each e_t is an e-value and the running product is a nonnegative supermartingale with unit initial value; Ville's inequality bounds $\mathbb{P}(\sup_t E_t \geq 1/\alpha) \leq \alpha$. $\square$

A.5 Residual inheritance (Prop. 7)

If $\hat{f}$ is equivariant then $r = y - \hat{f}(x, u)$ transforms in the same representation as y, so H_0 for Z implies H_0 for the residual element and Thm. 3 applies unchanged. If $\hat{f}$ is not equivariant, its defect $\delta_f = \hat{f} \circ \rho - \rho \circ \hat{f}$ appears in the residual as a deterministic antisymmetric offset, which enters $\Pi^- \mu$ and inflates the rejection rate under H_0; the H_0' construction removes it by differencing because the offset is common to calibration and monitoring windows. $\square$

A.6 Rolling contact (Prop. 8)

Writing the contact component of the propagation as $f_u^{d_i}(X) = Ru_i$ and expanding the group-affine identity in the $\mathrm{SE}_{2+m}(3)$ product, all three terms reduce to $RR_Y u_i$, so f_u is group-affine and the right-invariant error obeys an autonomous linear equation. Reading A off the world-constant parts, the velocity block inherits $[g]_\times$ from gravity while the contact block has no world-constant term — Ru_i is a pure transport of a known body vector — so $A[d_i, R] = 0$. Replacing u_i by $u_i + \delta u_i$ adds $R\,\mathrm{Cov}(\delta u_i)R^\top$ to the contact process noise and nothing else. $\square$